

Title

Prophylactic Negative Pressure Therapy Substantially Reduces Surgical Site Infection in Emergency Abdominal Surgery

Abstract

Ninety emergency surgical patients undergoing open abdominal procedures were randomized to either five days of prophylactic negative pressure wound closure ($n = 45$) or standard sterile dressing management ($n = 45$). Postoperative surgical site infection developed in 19 patients (42.2%) in the standard dressing group versus 8 patients (17.8%) in the negative pressure group ($p = 0.019$). Superficial infection incidence was significantly lower with prophylactic vacuum therapy (17.8% versus 37.8%, $p = 0.032$), while deep infection rates showed no meaningful difference (4.4% each group, $p = 1.0$). Additional benefits included accelerated wound healing progression and improved cosmetic appearance at 30-day assessment. One adverse event (skin irritation at vacuum interface) was documented in the negative pressure group. These findings demonstrate that prophylactic negative pressure wound therapy provides meaningful infection reduction without significant safety concerns, supporting its use in high-risk emergency surgical contexts.